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Supporting Information

Techno-economic comparison of fossil, renewable-carbon and recycling pathways for polypropylene

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S1 Purpose and reproducibility scope

This Supporting Information documents the individual process-step assumptions, the construction of the assessed polypropylene pathways, the aggregated pathway inputs passed to CaCoCa, the selected price scenarios, and the resulting annual cost outputs. It is intended to be read together with the manuscript and the archived model release.

The CaCoCa model starts from one aggregated parameter set per pathway. It does not reconstruct the route from unit processes. The individual process steps were combined before the CaCoCa calculation using pathway-specific yields, process scales, and allocation factors. The raw openLCA project files are not distributed because the working project contains legacy names and superseded intermediate values. Instead, the curated process-step inventory and machine-readable aggregation bridge provide the values and calculations required to reproduce the complete pathway aggregation. Accordingly:

- `process_step_data.csv` documents the individual process-step assumptions and sources;
- `aggregated_pathway_data.csv` reproduces the pathway-level inputs used by CaCoCa;
- `pathway_aggregation_calculations.csv` documents the process scale, allocation factor, and contribution of every process parameter to the aggregated pathway values;
- `publication_results.csv` provides annual results for all 14 pathways under the high and low carbon-price trajectories;
- `sensitivity_results_2045.csv` provides the one-at-a-time sensitivity ranges reported in the manuscript;
- `data_dictionary.csv` defines the columns in all machine-readable files.

All author-year citations in this document correspond to the reference list of the associated manuscript. The archived CSV files are the preferred source for machine-readable values.

S2 Functional unit, system boundary, and aggregation

The functional unit is 1 tonne of polypropylene pellets at the gate of the polymerisation plant after pelletisation. The process boundary is cradle-to-gate for the represented production chain. Downstream transport to plastics processors and the use phase are excluded. A hypothetical fossil end-of-life carbon component is reported separately for the two fossil-naphtha pathways and is not interpreted as a current production cost. For the purchased renewable-naphtha and renewable-propane cases, upstream feedstock production is represented by the feedstock price and is not added to the reported direct process emissions.

For each process chain, material and energy inputs, direct emissions, CAPEX, and additional OPEX are converted to the functional unit using the yield of each subsequent process step. Multi-output processes are allocated as described in Section S3. The file `pathway_aggregation_calculations.csv` records the effective process scale, allocation factor, native process value, pathway contribution, and corresponding CaCoCa aggregate for every parameter. The resulting pathway parameters are imported into CaCoCa as one aggregated record per pathway.

S3 Allocation of multi-output processes

Material feedstocks are allocated by mass. Energy demand and process emissions are allocated according to the share of the relevant product in the high-value-chemical product

slate. CAPEX and additional OPEX are allocated economically. Under the product shares and price assumptions below, propylene receives 28.2% of the economically allocated CAPEX and OPEX.

Product	Mass share	Reported price range	Assumed price (EUR/t)	Economic allocation share
Ethylene	0.245	USD 790-870/t	713.80	0.314
Propylene	0.196	USD 860-1,002/t	800.66	0.282
Butenes and butadiene	0.108	approximately USD 790-805/t	685.85	0.133
BTX aromatics	0.138	USD 800-923/t	720.39	0.179
C5+ products	0.073	approximately USD 665/t	700.04	0.092

The allocation factors are also provided in `allocation_factors.csv`. The product-price assumptions are author calculations and should not be interpreted as a current market-price dataset.

S4 Process-step inventory

The following tables list the unit-process or integrated-process parameters used to construct the pathways. The “assumption status” column distinguishes literature values from author calculations and explicit author assumptions. Values are shown on their native process basis unless they are explicitly identified as converted pathway contributions on the PP functional unit. Together with `pathway_aggregation_calculations.csv`, the tables provide the information needed to reproduce the conversion to one tonne of PP without access to the raw openLCA project files.

S4.1 Conventional steam cracker

Parameter	Value	Unit	Source / basis	Assumption status
HVC yield	0.760	t HVC/t cracker feed	FutureCamp and Dechema (2019)	Literature-based
Propylene yield	0.196	t propylene/t cracker feed	FutureCamp and Dechema (2019)	Literature-based
CAPEX	2,250.00	EUR/t annual HVC capacity	FutureCamp and Dechema (2019); author conversion	Literature-based
Converted CAPEX basis	1,710.00	EUR/t annual feed capacity	Author conversion from HVC basis	Author calculation
Natural-gas demand	16.50	GJ/t HVC	FutureCamp and Dechema (2019)	Literature-based
Natural-gas demand	4.580	MWh/t HVC	Author unit conversion	Author calculation
Electricity demand	1.000	GJ/t HVC	FutureCamp and Dechema (2019)	Literature-based
Electricity demand	0.280	MWh/t HVC	Author unit conversion	Author calculation
Direct process emissions	1.350	t CO ₂ /t propylene	Alsabri et al. (2021)	Literature-based
Additional OPEX	43.75	EUR/t cracker feed	2.5% of allocated CAPEX	Author assumption

S4.2 Electric steam cracker

Parameter	Value	Unit	Source / basis	Assumption status
HVC yield	0.760	t HVC/t cracker feed	FutureCamp and Dechema (2019); same material conversion as conventional cracker	Literature-based with author assumption
Propylene yield	0.196	t propylene/t cracker feed	FutureCamp and Dechema (2019); same material conversion as conventional cracker	Literature-based with author assumption
CAPEX	2,250.00	EUR/t annual HVC capacity	FutureCamp and Dechema (2019); greenfield assumption of identical total cracker CAPEX	Author assumption
Natural-gas demand	0.000	GJ/t HVC	Author calculation: fossil-fired heat fully replaced by electricity	Author assumption
Electricity demand	17.50	GJ/t HVC	FutureCamp and Dechema (2019); total cracker energy supplied electrically	Literature-based with author conversion
Electricity demand	4.860	MWh/t HVC	Author unit conversion	Author calculation
Direct combustion emissions	0.000	t CO ₂ /t propylene	Author assumption for fully electric heat supply; polymerisation emissions remain separate	Author assumption
Additional OPEX	43.75	EUR/t cracker feed	2.5% of allocated CAPEX	Author assumption

The electric-cracker configuration represents a greenfield comparison. It substitutes electricity for fossil-fired process heat but retains the same material conversion and carbon feedstock. The renewable-naphtha pathway applies this cracker and polymerisation inventory to purchased renewable naphtha; cracker-heat emissions are therefore not assigned.

S4.3 Polymerisation

Parameter	Value	Unit	Source / basis	Assumption status
PP yield	0.985	t PP/t propylene	Author assumption; Equivalent to 1.015 t propylene input per tonne PP	Author assumption
CAPEX	200.00	EUR/t annual PP capacity	Rueda-Durán et al. (2018); rounded author assumption	Literature-based with author adjustment
Propylene input	1.015	t propylene/t PP	Author conversion from 98.5% yield	Author calculation
Electricity demand	0.230	MWh/t PP	Volk et al. (2021)	Literature-based
Direct process emissions	0.200	t CO ₂ /t PP	Alsabri et al. (2021); final harmonised value used in the pathway aggregation	Literature-based with author harmonisation
Additional OPEX	5.000	EUR/t PP	2.5% of the assumed CAPEX of EUR 200/t annual PP capacity	Author calculation based on assumed CAPEX

S4.4 Plastic-waste pyrolysis

Parameter	Value	Unit	Source / basis	Assumption status
Pyrolysis-oil yield	0.600	t pyrolysis oil/t PCW	Biessey et al. (2023); Stallkamp (2023); Stallkamp et al. (2024)	Literature-based central estimate
CAPEX	100.00	EUR/t annual PCW capacity	Agora Energiewende and Wuppertal Institut (2019)	Literature-based
PCW input	1.667	t PCW/t pyrolysis oil	Author conversion from yield	Author calculation
Electricity demand	0.233	MWh/t PCW	Ausfelder et al. (2026)	Literature-based

Direct process emissions	0.440	t CO ₂ /t PCW	Volk et al. (2021)	Literature-based
Additional OPEX	2.000	EUR/t PCW	Agora Energiewende and Wuppertal Institut (2019)	Literature-based

S4.5 Pyrolysis-oil upgrading

Parameter	Value	Unit	Source / basis	Assumption status
Upgraded-oil yield	0.939	t upgraded oil/t pyrolysis oil	Möck et al. (2022)	Literature-based
CAPEX	456.00	EUR/t annual pyrolysis-oil capacity	Letoffet et al. (2024); Sphera (2022)	Literature-based
Pyrolysis-oil input	1.065	t pyrolysis oil/t upgraded oil	Author conversion from yield	Author calculation
Electricity demand	0.098	MWh/t pyrolysis oil	Letoffet et al. (2024)	Literature-based
Direct process emissions	0.187	t CO ₂ /t pyrolysis oil	Letoffet et al. (2024)	Literature-based
Hydrogen demand	10.00	kg H ₂ /t pyrolysis oil	Dechema et al. (2021); Sphera (2022)	Literature-based
Additional OPEX	11.40	EUR/t pyrolysis oil	2.5% of CAPEX	Author assumption

S4.6 Steam cracking of upgraded pyrolysis oil

Parameter	Value	Unit	Source / basis	Assumption status
HVC yield	0.608	t HVC/t cracker feed	80% of the naphtha-based HVC yield of 0.760	Author assumption
Propylene yield	0.1568	t propylene/t cracker feed	80% of the naphtha-based propylene yield of 0.196	Author assumption
CAPEX	2,250.00	EUR/t annual HVC capacity	FutureCamp and Dechema (2019); author transfer assumption	Literature-based with author assumption
Natural-gas demand	16.50	GJ/t HVC	FutureCamp and Dechema (2019)	Literature-based
Electricity demand	1.000	GJ/t HVC	FutureCamp and Dechema (2019)	Literature-based
Direct process emissions	1.350	t CO ₂ /t propylene	Alsabri et al. (2021); transferred to upgraded-oil route	Literature-based with author assumption
Additional OPEX	85.50	EUR/t cracker feed	5% of allocated CAPEX to reflect more demanding feedstock	Author assumption

The lower propylene yield and higher OPEX for upgraded pyrolysis oil are conservative author assumptions because pure-feedstock steam-cracking evidence remains limited.

S4.7 Plastic-waste hydrocracking

Parameter	Value	Unit	Source / basis	Assumption status
Liquid-product yield	0.900	t naphtha substitute/t PCW	Author assumption informed by Dechema et al. (2021) and Olafasakin et al. (2023)	Author assumption
CAPEX	2,500.00	EUR/t annual PCW capacity	Olafasakin et al. (2023); Transferred from related thermochemical process evidence	Literature-based
PCW input	1.111	t PCW/t naphtha substitute	Author conversion from yield	Author calculation
Electricity demand	0.400	MWh/t PCW	Author assumption	Author assumption
Direct process emissions	0.702	t CO ₂ /t PCW	Olafasakin et al. (2023)	Literature-based

Hydrogen demand	30.00	kg H ₂ /t PCW	Author assumption within range reported by Dechema et al. (2021)	Author assumption
Additional OPEX	125.00	EUR/t PCW	5% of CAPEX	Author assumption

Hydrocracking values have a comparatively high uncertainty because the available evidence is not fully specific to post-consumer polypropylene feedstock.

S4.8 Plastic-waste gasification and Fischer-Tropsch synthesis

Parameter	Value	Unit	Source / basis	Assumption status
FT-naphtha yield	0.603	t FT naphtha/t PCW	Ketting (2024)	Literature-based
CAPEX	2,505.00	EUR/t annual PCW capacity	Ketting (2024)	Literature-based
Electricity demand	0.100	MWh/t PCW	Ketting (2024)	Literature-based
Direct process emissions	1.190	t CO ₂ /t PCW	Author conversion from Lee et al. (2023)	Author calculation
Additional OPEX	207.79	EUR/t PCW	Author assumption used in the process-chain aggregation	Author assumption

S4.9 Mechanical recycling

Parameter	Value	Unit	Source / basis	Assumption status
rPP yield	0.850	t rPP/t PCW	Nordahl et al. (2023)	Literature-based
CAPEX	95.00	EUR/t annual PCW capacity	Garcia-Gutierrez et al. (2023)	Literature-based
PCW input	1.180	t PCW/t rPP	Author conversion from yield	Author calculation
Electricity demand	0.492	MWh/t PCW	Midpoint of Larrain et al. (2021) and Nordahl et al. (2023)	Literature-based central estimate
Direct process emissions	0.400	t CO ₂ /t PCW	Nordahl et al. (2023)	Literature-based
Additional OPEX	157.40	EUR/t PCW	Garcia-Gutierrez et al. (2023); Reported value may include electricity; retained as applied in the model basis.	Literature-based

S4.10 CO₂ capture by amine scrubbing

Parameter	Value	Unit	Source / basis	Assumption status
Capture rate	90.00	%	Sollai et al. (2023)	Literature-based
CAPEX	615.70	EUR/t annual CO ₂ capture capacity	Sollai et al. (2023)	Literature-based
Electricity demand	0.972	MWh/t captured CO ₂	Author assumption based on the process configuration in Sollai et al. (2023)	Author assumption
Residual direct emissions	0.110	t CO ₂ /t captured CO ₂	Author calculation based on 90% capture rate	Author calculation
Additional OPEX	15.39	EUR/t captured CO ₂	2.5% of CAPEX	Author assumption

The capture-electricity value is retained as an author assumption based on the selected amine-scrubbing configuration.

S4.11 Reverse water-gas shift and methanol synthesis

Parameter	Value	Unit	Source / basis	Assumption status
Methanol yield	0.605	t methanol/t (H ₂ + CO ₂)	Sollai et al. (2023)	Literature-based
CAPEX	280.82	EUR/t annual methanol capacity	Sollai et al. (2023)	Literature-based
Electricity demand	0.207	MWh/t methanol	Sollai et al. (2023)	Literature-based
CO ₂ demand	1.446	t CO ₂ /t methanol	Sollai et al. (2023)	Literature-based
Hydrogen demand	0.208	t H ₂ /t methanol	Sollai et al. (2023)	Literature-based
Direct process emissions	0.050	t CO ₂ /t methanol	Sollai et al. (2023)	Literature-based
Water coproduct	0.654	t H ₂ O/t methanol	Sollai et al. (2023)	Literature-based
Additional OPEX	7.000	EUR/t methanol	2.5% of CAPEX	Author assumption

S4.12 Methanol-to-propylene

Parameter	Value	Unit	Source / basis	Assumption status
HVC yield	0.460	t HVC/t methanol	Markowitsch et al. (2023); product slate excluding water	Literature-based
CAPEX	1,106.75	EUR/t annual methanol input	Markowitsch et al. (2023); inflation and currency conversion	Literature-based with author conversion
Electricity demand	3.060	MWh/t methanol	Markowitsch et al. (2023); author conversion	Literature-based with author conversion
Additional OPEX	27.65	EUR/t methanol	2.5% of CAPEX	Author assumption

S4.13 Reverse water-gas shift and Fischer-Tropsch synthesis

Parameter	Value	Unit	Source / basis	Assumption status
FT-naphtha yield	0.265	t FT naphtha/t (H ₂ + CO ₂)	Markowitsch et al. (2023)	Literature-based
CAPEX	297.89	EUR/t annual FT-naphtha capacity	Author scaling based on Markowitsch et al. (2023)	Author calculation
Electricity demand	0.239	MWh/t FT naphtha	Author assumption for compression demand	Author assumption
CO ₂ demand	3.307	t CO ₂ /t FT naphtha	Markowitsch et al. (2023)	Literature-based
Hydrogen demand	461.90	kg H ₂ /t FT naphtha	Markowitsch et al. (2023)	Literature-based
Direct process emissions	0.016	t CO ₂ /t FT naphtha	Markowitsch et al. (2023)	Literature-based
Water coproduct	2.750	t H ₂ O/t FT naphtha	Markowitsch et al. (2023)	Literature-based
Additional OPEX	7.440	EUR/t FT naphtha	2.5% of CAPEX	Author assumption

S4.14 Biomethanol production

Parameter	Value	Unit	Source / basis	Assumption status
Methanol yield	0.385	t methanol/t dry biomass	Bazzanella and Ausfelder (2017)	Literature-based

CAPEX	1,360.00	EUR/t annual biomethanol capacity	Author scaling from Bazzanella and Ausfelder (2017), Markowitsch et al. (2023), and Sollai et al. (2023)	Author calculation
Electricity demand	0.167	MWh/t biomethanol	Bazzanella and Ausfelder (2017)	Literature-based
Additional OPEX	34.00	EUR/t biomethanol	2.5% of CAPEX	Author assumption

S4.15 Conventional propane dehydrogenation (PDH)

Parameter	Value	Unit	Source / basis	Assumption status
Renewable propane input	1.194	t renewable propane/t PP	Agarwal (2018); author scaling to the PP functional unit	Literature-based with author conversion
Electricity demand, PDH block	0.500	MWh/t PP	Agarwal (2018); author conversion	Literature-based with author conversion
Reported fixed CAPEX	990.00	USD_2016/t annual propylene capacity	Agarwal (2018)	Literature-based
Converted PDH CAPEX contribution	1,415.00	EUR/t annual PP capacity	Agarwal (2018); author currency, price-year and functional-unit conversion	Author calculation
Reported non-feedstock OPEX	76.10	USD_2016/t propylene	Agarwal (2018)	Literature-based
Converted PDH OPEX contribution	88.00	EUR/t PP	Agarwal (2018); author currency, price-year and functional-unit conversion	Author calculation
Hydrogen coproduct	0.0443	t H ₂ /t PP	Agarwal (2018); author scaling to the PP functional unit; no cost credit applied	Literature-based with author conversion

S4.16 PDH utility integration and off-gas heat recovery

Parameter	Value	Unit	Source / basis	Assumption status
Low-pressure steam demand	3.333	MWh _{th} /t PP	Agarwal (2018); author unit and functional-unit conversion	Literature-based with author conversion
Fired-heat requirement	2.237	MWh _{th} /t PP	Agarwal (2018); author unit and functional-unit conversion	Literature-based with author conversion
Final purchased natural-gas demand	4.878	MWh/t PP	Agarwal (2018); author utility integration and off-gas heat-recovery calculation	Author calculation
Direct process emissions contribution	0.980	t CO ₂ /t PP	Agarwal (2018); author conversion and boundary treatment	Author calculation

The net natural-gas input retains the external utility demand after energetic recovery of renewable PDH off-gas. The route receives no hydrogen coproduct credit. Including the common polymerisation step gives 0.730 MWh electricity/t PP, 1.180 t CO₂/t PP, CAPEX of EUR 1,615/t annual PP capacity, and additional OPEX of EUR 93/t PP.

S5 Composition of the 14 assessed pathways

Model project name	Reader-facing pathway	Process chain
Steamcracker Naphtha	Fossil naphtha via conventional steam cracking	Conventional steam cracker -> Polymerisation
E-Steamcracker Naphtha	Fossil naphtha via electric steam cracking	Electric steam cracker -> Polymerisation
Pyrolyse	PCW via pyrolysis and conventional steam cracking	Plastic-waste pyrolysis -> Pyrolysis-oil upgrading -> Steam cracking of upgraded pyrolysis oil -> Polymerisation
Pyrolyse E-Steamcracker	PCW via pyrolysis and electric steam cracking	Plastic-waste pyrolysis -> Pyrolysis-oil upgrading -> Electric steam cracking -> Polymerisation
Hydrocracking	PCW via hydrocracking and conventional steam cracking	Plastic-waste hydrocracking -> Conventional steam cracker -> Polymerisation
Hydrocracking E-Steamcracker	PCW via hydrocracking and electric steam cracking	Plastic-waste hydrocracking -> Electric steam cracker -> Polymerisation
Gasifizierung-FT	PCW via gasification-FT and conventional steam cracking	Plastic-waste gasification and Fischer-Tropsch synthesis -> Conventional steam cracker -> Polymerisation
Gasifizierung-FT E-Steamcracker	PCW via gasification-FT and electric steam cracking	Plastic-waste gasification and Fischer-Tropsch synthesis -> Electric steam cracker -> Polymerisation
Mechanisches Recycling	PCW via mechanical recycling	Mechanical recycling
Methanolroute	Captured CO ₂ and H ₂ via methanol-to-propylene	CO ₂ capture by amine scrubbing -> Reverse water-gas shift and methanol synthesis -> Methanol-to-propylene -> Polymerisation
Fischer-Tropsch	Captured CO ₂ and H ₂ via Fischer-Tropsch and electric steam cracking	CO ₂ capture by amine scrubbing -> Reverse water-gas shift and Fischer-Tropsch synthesis -> Electric steam cracker -> Polymerisation
Biomethanol	Dry biomass via biomethanol-to-propylene	Biomethanol production -> Methanol-to-propylene -> Polymerisation
Renewable Naphtha	Renewable naphtha via electric steam cracking	Electric steam cracker → Polymerisation
Renewable Propane	Renewable propane via conventional PDH	Conventional propane dehydrogenation (PDH) → PDH utility integration and off-gas heat recovery → Polymerisation

S6 Aggregated pathway inputs used in CaCoCa

Table S6.1 reproduces the aggregated pathway inputs used by the model. These values are the authoritative CaCoCa inputs. They are also available in aggregated_pathway_data.csv and in data/tech/basic/polymers.csv in the model repository.

Pathway	CAPEX (EUR/t annual PP capacity)	Natural gas (MWh/t PP)	Electricity (MWh/t PP)	Direct emissions (t CO ₂ /t PP)	Additional OPEX (EUR/t PP)	Feedstock demands
Fossil naphtha, conventional	2,710.79	4.650	0.510	1.570	67.77	Naphtha: 1.015 t/t PP; CO ₂ EoL: 3.140 t/t PP
Fossil naphtha, electric	2,710.79	0.000	5.160	0.200	67.77	Naphtha: 1.015 t/t PP; CO ₂ EoL: 3.140 t/t PP
Pyrolysis, conventional	4,011.55	5.810	1.111	2.908	177.85	PCW: 1.801 t/t PP; Hydrogen: 10.809 kg/t PP
Pyrolysis, electric	4,011.55	0.000	6.922	1.195	177.85	PCW: 1.800 t/t PP; Hydrogen: 10.809 kg/t PP
Hydrocracking, conventional	5,530.23	4.650	0.970	2.360	208.74	PCW: 1.130 t/t PP; Hydrogen: 33.830 kg/t PP
Hydrocracking, electric	5,530.23	0.000	5.610	0.990	208.74	PCW: 1.130 t/t PP; Hydrogen: 33.830 kg/t PP
Gasification-FT, conventional	6,927.33	4.650	0.680	3.580	417.53	PCW: 1.680 t/t PP
Gasification-FT, electric	6,927.33	0.000	5.330	2.210	417.53	PCW: 1.680 t/t PP
Mechanical recycling	111.76	0.000	0.579	0.471	185.18	PCW: 1.180 t/t PP
CO ₂ /H ₂ methanol route	5,627.57	0.000	11.350	0.319	140.58	Hydrogen: 495.590 kg/t PP
CO ₂ /H ₂ FT route	5,080.04	0.000	8.668	0.216	126.98	Hydrogen: 468.830 kg/t PP
Biomethanol route	6,077.00	0.000	7.910	0.200	151.89	Biomass: 6.198 t/t PP
Renewable naphtha, electric cracker	2,710.79	0.000	5.160	0.200	67.77	Renewable naphtha: 1.015 t/t PP
Renewable propane, conventional PDH	1,615.00	4.878	0.730	1.180	93.00	Renewable propane: 1.194 t/t PP

Hydrogen is reported in kg/t PP; other material feedstocks are reported in t/t PP. The hypothetical “CO₂ EoL” component is a feedstock-like model component used only for the two fossil-naphtha pathways. Renewable naphtha and renewable propane are represented as purchased feedstocks; the route parameters above reproduce the final CaCoCa input records.

S7 Price scenarios and sensitivity design

Electricity, natural gas, hydrogen, and carbon prices vary annually. Material-feedstock prices remain constant in the baseline. The high and low carbon-price cases use the Ariadne KN2045_Mix and ExPol trajectories, respectively. Direct process CO₂ and the hypothetical fossil end-of-life component use the same selected trajectory within each scenario.

Component	Scenario	Year	Value	Unit	Source
Electricity	Industry70-150GWh	2025	110.80	€/MWh	Treibhausgas-Projektionen 2026
Electricity	Industry70-150GWh	2035	80.20	€/MWh	Treibhausgas-Projektionen 2026
Electricity	Industry70-150GWh	2045	69.80	€/MWh	Treibhausgas-Projektionen 2026
Natural Gas	Industry>4000GJ	2025	55.73	€/MWh	Treibhausgas-Projektionen 2026
Natural Gas	Industry>4000GJ	2035	28.22	€/MWh	Treibhausgas-Projektionen 2026
Natural Gas	Industry>4000GJ	2045	26.99	€/MWh	Treibhausgas-Projektionen 2026
Hydrogen	IndustryLarge	2025	6.19	€/kg	Treibhausgas-Projektionen 2026
Hydrogen	IndustryLarge	2035	5.06	€/kg	Treibhausgas-Projektionen 2026
Hydrogen	IndustryLarge	2045	4.44	€/kg	Treibhausgas-Projektionen 2026
Naphtha	Constant baseline	2025	600.00	€/t	Rounded conservative market proxy; BASF (2026) reports 2025 average naphtha price of 567 USD/t
Naphtha	Constant baseline	2035	600.00	€/t	Rounded conservative market proxy; BASF (2026) reports 2025 average naphtha price of 567 USD/t
Naphtha	Constant baseline	2045	600.00	€/t	Rounded conservative market proxy; BASF (2026) reports 2025 average naphtha price of 567 USD/t
PCW	Constant baseline	2025	125.00	€/t	Author assumption based on Cimpan et al. (2016) and Garcia-Gutierrez et al. (2023)
PCW	Constant baseline	2035	125.00	€/t	Author assumption based on Cimpan et al. (2016) and Garcia-Gutierrez et al. (2023)
PCW	Constant baseline	2045	125.00	€/t	Author assumption based on Cimpan et al. (2016) and Garcia-Gutierrez et al. (2023)
Biomass	Constant baseline	2025	71.20	€/t	Haro et al. (2013), dry biomass basis
Biomass	Constant baseline	2035	71.20	€/t	Haro et al. (2013), dry biomass basis
Biomass	Constant baseline	2045	71.20	€/t	Haro et al. (2013), dry biomass basis
Renewable naphtha	Constant baseline	2025	1,350	€/t	Author assumption for purchased renewable naphtha; Revision 2 baseline
Renewable naphtha	Constant baseline	2035	1,350	€/t	Author assumption for purchased renewable naphtha; Revision 2 baseline
Renewable naphtha	Constant baseline	2045	1,350	€/t	Author assumption for purchased renewable naphtha; Revision 2 baseline
Renewable propane	Constant baseline	2025	1,200	€/t	Author assumption for purchased renewable propane; Revision 2 baseline
Renewable propane	Constant baseline	2035	1,200	€/t	Author assumption for purchased renewable propane; Revision 2 baseline
Renewable propane	Constant baseline	2045	1,200	€/t	Author assumption for purchased renewable propane; Revision 2 baseline

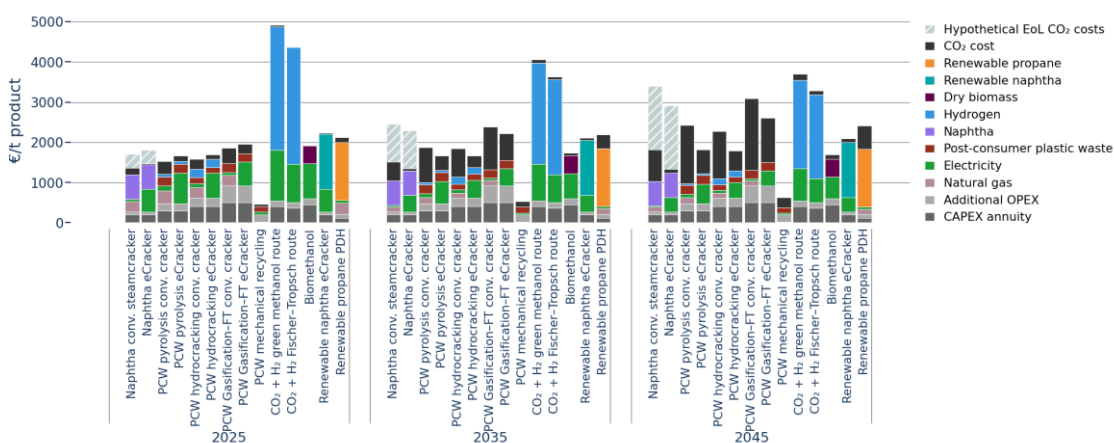
CO ₂	Ariadne-Indikation-KN2045_Mix	2025	108.84	€/t	Ariadne
CO ₂	Ariadne-Indikation-KN2045_Mix	2035	300.00	€/t	Ariadne
CO ₂	Ariadne-Indikation-KN2045_Mix	2045	500.00	€/t	Ariadne
CO ₂	Ariadne-Indikation-ExPol	2025	80.27	€/t	Ariadne
CO ₂	Ariadne-Indikation-ExPol	2035	140.00	€/t	Ariadne
CO ₂	Ariadne-Indikation-ExPol	2045	200.00	€/t	Ariadne
CO ₂ EoL	Ariadne-Indikation-KN2045_Mix	2025	108.84	€/t	Ariadne
CO ₂ EoL	Ariadne-Indikation-KN2045_Mix	2035	300.00	€/t	Ariadne
CO ₂ EoL	Ariadne-Indikation-KN2045_Mix	2045	500.00	€/t	Ariadne
CO ₂ EoL	Ariadne-Indikation-ExPol	2025	80.27	€/t	Ariadne
CO ₂ EoL	Ariadne-Indikation-ExPol	2035	140.00	€/t	Ariadne
CO ₂ EoL	Ariadne-Indikation-ExPol	2045	200.00	€/t	Ariadne

The one-at-a-time sensitivity analysis varies electricity, natural gas, hydrogen, PCW, dry biomass, renewable naphtha, and renewable propane prices separately by $\pm 20\%$ while holding all other parameters constant. Carbon-price uncertainty is represented by the difference between the low and high exogenous trajectories rather than by a $\pm 20\%$ variation.

S8 Publication results and sensitivity outputs

The complete annual results for 2025–2045 are provided in publication_results.csv. The tables below provide selected values for 2025, 2035, and 2045. ‘Current regulatory cost’ includes direct process CO₂ costs but excludes the hypothetical fossil end-of-life component; ‘analytical total’ includes both direct process CO₂ costs and the separate hypothetical end-of-life component.

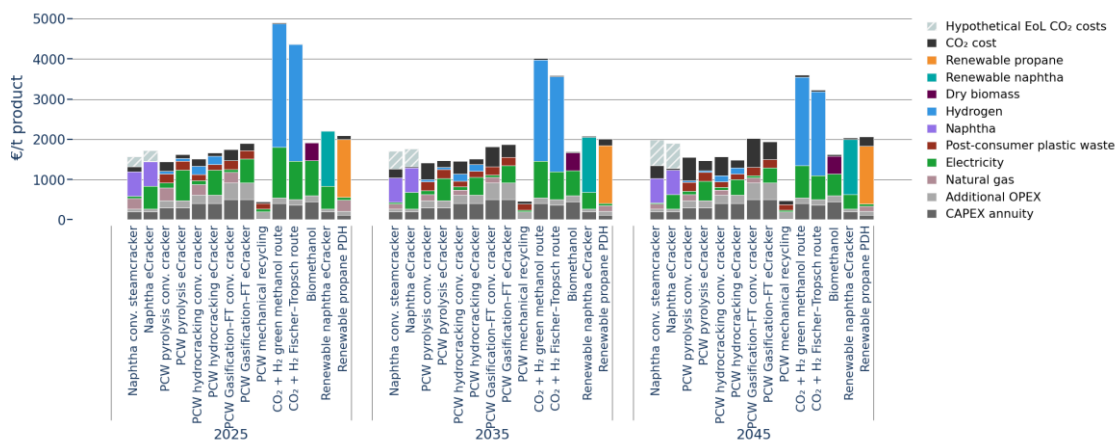
S8.1 High carbon-price trajectory



Pathway	2025 current	2025 total	2035 current	2035 total	2045 current	2045 total
Fossil naphtha via conventional steam cracking	1,360.24	1,701.99	1,516.83	2,458.83	1,819.81	3,389.81
Fossil naphtha via electric steam cracking	1,467.20	1,808.96	1,347.54	2,289.54	1,333.87	2,903.87
PCW via pyrolysis and conventional steam cracking	1,524.75	1,524.75	1,874.59	1,874.59	2,430.73	2,430.73
PCW via pyrolysis and electric steam cracking	1,658.21	1,658.21	1,662.59	1,662.59	1,822.80	1,822.80
PCW via hydrocracking and conventional steam cracking	1,584.77	1,584.77	1,840.05	1,840.05	2,275.07	2,275.07
PCW via hydrocracking and electric steam cracking	1,690.62	1,690.62	1,669.96	1,669.96	1,788.44	1,788.44
PCW via gasification-FT and conventional steam cracking	1,854.93	1,854.93	2,390.55	2,390.55	3,093.76	3,093.76
PCW via gasification-FT and electric steam cracking	1,961.89	1,961.89	2,221.26	2,221.26	2,607.83	2,607.83
PCW via mechanical recycling	459.60	459.60	531.92	531.92	620.10	620.10
Captured CO ₂ and H ₂ via methanol-to-propylene	4,911.20	4,911.20	4,064.56	4,064.56	3,700.16	3,700.16
Captured CO ₂ and H ₂ via Fischer-Tropsch and electric steam cracking	4,383.70	4,383.70	3,629.65	3,629.65	3,289.26	3,289.26
Dry biomass via biomethanol-to-propylene	1,932.87	1,932.87	1,729.06	1,729.06	1,686.79	1,686.79
Renewable naphtha via electric steam cracking	2,228.45	2,228.45	2,108.79	2,108.79	2,095.12	2,095.12
Renewable propane via conventional PDH	2,124.29	2,124.29	2,193.33	2,193.33	2,415.74	2,415.74

Costs in EUR/t PP. For pathways without a hypothetical end-of-life component, current regulatory cost and analytical total are identical.

S8.2 Low carbon-price trajectory



Pathway	2025 current	2025 total	2035 current	2035 total	2045 current	2045 total
Fossil naphtha via conventional steam cracking	1,315.38	1,567.43	1,265.63	1,705.23	1,348.81	1,976.81
Fossil naphtha via electric steam cracking	1,461.49	1,713.54	1,315.54	1,755.14	1,273.87	1,901.87
PCW via pyrolysis and conventional steam cracking	1,441.67	1,441.67	1,409.31	1,409.31	1,558.33	1,558.33
PCW via pyrolysis and electric steam cracking	1,624.08	1,624.08	1,471.41	1,471.41	1,464.35	1,464.35
PCW via hydrocracking and conventional steam cracking	1,517.34	1,517.34	1,462.45	1,462.45	1,567.07	1,567.07
PCW via hydrocracking and electric steam cracking	1,662.34	1,662.34	1,511.56	1,511.56	1,491.44	1,491.44
PCW via gasification-FT and conventional steam cracking	1,752.65	1,752.65	1,817.75	1,817.75	2,019.76	2,019.76
PCW via gasification-FT and electric steam cracking	1,898.75	1,898.75	1,867.66	1,867.66	1,944.83	1,944.83
PCW via mechanical recycling	446.15	446.15	456.56	456.56	478.80	478.80
Captured CO ₂ and H ₂ via methanol-to-propylene	4,902.09	4,902.09	4,013.49	4,013.49	3,604.42	3,604.42
Captured CO ₂ and H ₂ via Fischer-Tropsch and electric steam cracking	4,377.53	4,377.53	3,595.10	3,595.10	3,224.48	3,224.48
Dry biomass via biomethanol-to-propylene	1,927.16	1,927.16	1,697.06	1,697.06	1,626.79	1,626.79
Renewable naphtha via electric steam cracking	2,222.74	2,222.74	2,076.79	2,076.79	2,035.12	2,035.12
Renewable propane via conventional PDH	2,090.58	2,090.58	2,004.53	2,004.53	2,061.74	2,061.74

Costs in EUR/t PP. For pathways without a hypothetical end-of-life component, current regulatory cost and analytical total are identical.

S8.3 Sensitivity results for 2045

Pathway	Electricity	Natural gas	Hydrogen	CO ₂ low–high	PCW	Material feedstock
Fossil naphtha, conventional	14	50	–	1413	–	–
Fossil naphtha, electric	144	–	–	1002	–	–
Pyrolysis, conventional	31	63	19	872	90	–
Pyrolysis, electric	193	–	19	358	90	–
Hydrocracking, conventional	27	50	60	708	56	–
Hydrocracking, electric	157	–	60	297	56	–
Gasification–FT, conventional	19	50	–	1074	84	–
Gasification–FT, electric	149	–	–	663	84	–
Mechanical recycling	16	–	–	141	59	–
CO ₂ /H ₂ methanol route	317	–	880	96	–	–
CO ₂ /H ₂ FT route	242	–	832	65	–	–
Biomethanol route	221	–	–	60	–	Dry biomass: 177
Renewable naphtha, electric cracker	144	–	–	60	–	Renewable naphtha: 548
Renewable propane, conventional PDH	20	53	–	354	–	Renewable propane: 573

Values are the full cost range in EUR/t PP between the –20% and +20% price cases. The CO₂ column is the difference between the low and high carbon-price trajectories. For the fossil-naphtha pathways, it includes the separately attributed hypothetical end-of-life factor.

S9 Machine-readable files and data dictionary

File	Content
process_step_data.csv	Long-format process-step inventory, including sources and assumption status
pathway_composition.csv	Mapping of the 14 model projects to reader-facing names and ordered process chains
aggregated_pathway_data.csv	Aggregated pathway inputs corresponding to polymers.csv
allocation_factors.csv	Economic allocation factors for cracker coproducts
scenario_inputs.csv	Selected annual price trajectories for energy, feedstocks, direct CO ₂ , and hypothetical EoL CO ₂
publication_results.csv	Annual model results for all 14 pathways under high and low carbon-price trajectories
sensitivity_results_2045.csv	One-at-a-time and carbon-scenario cost ranges used in the manuscript
data_dictionary.csv	Definition, unit, and interpretation of every column in the machine-readable files
Supporting_Information_PP_pathways_data.xlsx	Workbook containing the above datasets as separate sheets
pathway_aggregation_calculations.csv	Process-step scaling, allocation factors, and parameter contributions to each aggregated pathway value; enables independent reconstruction without the raw openLCA project.

The repository README explains the relationship between the unit-process data, the aggregation bridge, aggregated pathway inputs, scenario files, and CaCoCa output. The Zenodo release contains the same file versions as the resubmitted manuscript. The raw openLCA project files are not part of the archive; process_step_data.csv and pathway_aggregation_calculations.csv provide a curated and independently usable replacement for reproducibility.

S10 Interpretation and limitations

- The process-step data combine commercial, demonstration-scale, and conceptual mature-plant estimates. No systematic first-of-a-kind adjustment or technology-specific learning curve is applied.
- The aggregated pathway dataset is authoritative for reproducing the CaCoCa calculation. The individual process-step values, scaling factors, allocation factors, and contributions to every aggregate are documented in `process_step_data.csv` and `pathway_aggregation_calculations.csv`. Minor differences caused only by displayed rounding do not affect the model inputs.
- Material-feedstock prices are constant in the baseline because consistent pathway-specific long-term trajectories were unavailable.
- The common PCW price does not capture differences in waste quality, composition, or pre-treatment requirements between recycling technologies.
- The renewable-naphtha and renewable-propane cases model purchased feedstocks at constant author-assumed prices rather than upstream production technologies; the $\pm 20\%$ sensitivities therefore represent procurement-price uncertainty.
- The conventional-PDH case retains external utility demand and renewable off-gas heat recovery but applies no hydrogen coproduct credit. Its direct-emissions value follows the process boundary used in the final manuscript.
- The emission values are aggregated direct process emissions used in the cost model, not a complete consequential life-cycle assessment.
- The hypothetical fossil end-of-life component is not a cost currently paid by polypropylene producers and is reported separately.